

## Benchmark Requirements/Metrics

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## Purpose

This document defines the requirements for local Large Language Model (LLM) benchmarking environment.

The purpose of the environment is to systematically evaluate open-weight LLMs across different parameter sizes and use cases, measuring their performance, resource consumption, output quality and reliability.

## Scope

The benchmark will be evaluated across 4 main use cases:

- Chat and General Reasoning (CGR)
- Retrieval-Augmented Generation (RAG)
- Classification
- Agentic Tool Use

The benchmarking will be done with mock data.

## Research Objective

The benchmark will provide sufficient evidence to answer the main question:

***“To what extent can locally hosted open-weight LLMs, running on limited hardware, support chat, retrieval, classification and agentic-tool use applications, and under what circumstances would additional hardware investment be justified.”***

## Functional Requirements

Functional Requirements describe what the benchmarking environment must be capable of performing,

| ID    | Requirement                                                                     | Priority |
|-------|---------------------------------------------------------------------------------|----------|
| FR-01 | The system must execute inference using locally hosted LLMs.                    | MUST     |
| FR-02 | The system must support benchmarking multiple models and parameter sizes.       | MUST     |
| FR-03 | The system must execute predefined benchmark prompts.                           | MUST     |
| FR-04 | The system must measure and record inference performance.                       | MUST     |
| FR-05 | The system must measure and record hardware resource consumption.               | MUST     |
| FR-06 | The system must store benchmark results in a structured local database.         | MUST     |
| FR-07 | The system must record model configuration and execution environment details.   | MUST     |
| FR-08 | The system must support repeated benchmark executions.                          | MUST     |
| FR-09 | The system must record model responses and execution errors.                    | MUST     |
| FR-10 | The system must evaluate results against predefined use-case-specific criteria. | MUST     |
| FR-11 | The system should support automated execution of complete benchmark suites.     | SHOULD   |

|       |                                                                                    |        |
|-------|------------------------------------------------------------------------------------|--------|
| FR-12 | The system should support exporting results to CSV or JSON.                        | SHOULD |
| FR-13 | The system should support comparing models across performance and quality metrics. | SHOULD |
| FR-14 | The system could provide a visualization dashboard for benchmark results.          | COULD  |
| FR-15 | The system could support configurable benchmark workloads and concurrency tests.   | COULD  |

## Non-functional Requirements

| ID     | Requirement                                                                                               | Priority |
|--------|-----------------------------------------------------------------------------------------------------------|----------|
| NFR-01 | Model inference must execute on the allocated ASUS hardware.                                              | Must     |
| NFR-02 | The benchmark must operate without external AI inference APIs.                                            | Must     |
| NFR-03 | Benchmark execution must be possible without internet access after initial setup.                         | Must     |
| NFR-04 | Benchmark datasets, model outputs, and measurements must remain stored locally during execution.          | Must     |
| NFR-05 | The environment must use consistent and documented configurations for comparable experiments.             | Must     |
| NFR-06 | Benchmark executions must be reproducible using recorded configurations, software versions, and datasets. | Must     |
| NFR-07 | Failed, timed-out, and crashed executions must be recorded rather than silently discarded.                | Must     |
| NFR-08 | The system should use containerization to simplify deployment and configuration management.               | Should   |
| NFR-09 | The system should support adding new models and benchmark categories without major architectural changes. | Should   |

|        |                                                                                              |      |
|--------|----------------------------------------------------------------------------------------------|------|
| NFR-10 | The benchmark database must support subsequent analysis without requiring tests to be rerun. | Must |
|--------|----------------------------------------------------------------------------------------------|------|

## Common / Base Metrics

| ID      | Metric                     | Unit   | What it tells you                                      |
|---------|----------------------------|--------|--------------------------------------------------------|
| BASE-01 | Model load time            | s      | How long the model takes to become ready               |
| BASE-02 | Time to First Token (TTFT) | s      | Initial responsiveness                                 |
| BASE-03 | Total latency              | s      | Total time to complete the request                     |
| BASE-04 | Input tokens               | tokens | Size of the input workload                             |
| BASE-05 | Output tokens              | tokens | Amount of generated output                             |
| BASE-06 | Prompt processing speed    | tok/s  | How quickly the model processes context                |
| BASE-07 | Generation speed           | tok/s  | How quickly it generates output                        |
| BASE-08 | Average CPU usage          | %      | CPU demand during execution                            |
| BASE-09 | Peak CPU usage             | %      | Maximum CPU demand                                     |
| BASE-10 | Average RAM usage          | GB     | Typical memory consumption                             |
| BASE-11 | Peak RAM usage             | GB     | Maximum memory requirement                             |
| BASE-12 | Swap usage                 | GB     | Whether the system is exceeding practical RAM capacity |
| BASE-13 | Model storage size         | GB     | Disk requirement                                       |
| BASE-14 | GPU utilization            | %      | GPU usage, if applicable                               |

|         |                        |           |                                                           |
|---------|------------------------|-----------|-----------------------------------------------------------|
| BASE-15 | VRAM usage             | GB        | GPU memory requirement, if applicable                     |
| BASE-16 | Execution success      | pass/fail | Whether the request completed                             |
| BASE-17 | Timeout                | yes/no    | Whether execution exceeded the allowed limit              |
| BASE-18 | Crash/OOM              | yes/no    | Whether the model caused a runtime/memory failure         |
| BASE-19 | Output format validity | %         | Whether the output conforms to the required format/schema |
| BASE-20 | Context length         | tokens    | How much context was supplied                             |
|         |                        |           |                                                           |

## Use-Case based Metrics

### Chat / General LLM

| ID      | Metric                           | Unit      | What it measures                                      |
|---------|----------------------------------|-----------|-------------------------------------------------------|
| CHAT-01 | Answer correctness               | score / % | Whether the answer is correct                         |
| CHAT-02 | Instruction following            | score / % | Whether the model followed the requested instructions |
| CHAT-03 | Relevance                        | score / % | Whether the answer actually addresses the question    |
| CHAT-04 | Factual accuracy                 | score / % | Whether factual claims are correct                    |
| CHAT-05 | Hallucination rate               | %         | Unsupported/incorrect claims                          |
| CHAT-06 | Structured output compliance     | %         | Whether requested JSON/format is valid                |
| CHAT-07 | Long-context quality degradation | % change  | Quality change as context grows                       |

|         |                      |   |                                                                   |
|---------|----------------------|---|-------------------------------------------------------------------|
| CHAT-08 | Response consistency | % | Whether repeated runs produce acceptable outcomes                 |
| CHAT-09 | Abstention accuracy  | % | Whether the model correctly says it doesn't know when appropriate |

## RAG

### Retrieval

| ID     | Metric                  | Unit  | What it measures                                   |
|--------|-------------------------|-------|----------------------------------------------------|
| RAG-01 | Recall-Top-K-Results    | %     | Whether the relevant document appears in the top K |
| RAG-02 | Precision-Top-K-Results | %     | How many retrieved documents are actually relevant |
| RAG-03 | MRR                     | score | Whether the relevant document is ranked highly     |
| RAG-04 | Retrieval latency       | ms/s  | Time required to retrieve context                  |

### Generation

| ID     | Metric              | Unit      | What it measures                                  |
|--------|---------------------|-----------|---------------------------------------------------|
| RAG-05 | Answer correctness  | % / score | Whether the final answer is correct               |
| RAG-06 | Groundedness        | % / score | Whether claims are supported by retrieved context |
| RAG-07 | Answer completeness | % / score | Whether all relevant                              |

|        |                        |   |                                                                             |
|--------|------------------------|---|-----------------------------------------------------------------------------|
|        |                        |   | information was included                                                    |
| RAG-08 | Unsupported claim rate | % | Claims not supported by retrieved information                               |
| RAG-09 | Citation accuracy      | % | Whether citations actually support the claims                               |
| RAG-10 | Abstention accuracy    | % | Whether the model avoids inventing answers when information isn't available |

## Classification

| ID     | Metric                     | Unit   | What it measures                                             |
|--------|----------------------------|--------|--------------------------------------------------------------|
| CLS-01 | Accuracy                   | %      | Overall correct classifications                              |
| CLS-02 | Precision                  | %      | Correct positive predictions                                 |
| CLS-03 | Recall                     | %      | Correctly identified actual cases                            |
| CLS-04 | Macro F1                   | score  | Overall balanced classification quality                      |
| CLS-05 | Per-class F1               | score  | Performance for each category                                |
| CLS-06 | Confusion matrix           | matrix | Which categories are confused                                |
| CLS-07 | Invalid output rate        | %      | Outputs outside allowed classes                              |
| CLS-08 | Classification consistency | %      | Stability across repeated runs                               |
| CLS-09 | Confidence/calibration     | score  | Whether confidence reflects actual correctness, if available |

## Agentic Tool-Use/Workflows

| ID      | Metric                                | Unit  | What it measures                                  |
|---------|---------------------------------------|-------|---------------------------------------------------|
| TOOL-01 | End-to-end task success               | %     | Whether the complete task was correctly completed |
| TOOL-02 | Tool selection accuracy               | %     | Whether the correct tool was selected             |
| TOOL-03 | Argument accuracy                     | %     | Whether correct parameters were generated         |
| TOOL-04 | Schema compliance                     | %     | Whether tool calls match the required schema      |
| TOOL-05 | Tool call count                       | count | Number of calls required                          |
| TOOL-06 | Unnecessary tool calls                | count | Inefficient actions                               |
| TOOL-07 | Multi-step success rate               | %     | Ability to complete multi-step workflows          |
| TOOL-08 | Error recovery rate                   | %     | Ability to recover from tool/API failures         |
| TOOL-09 | Task completion latency               | s     | Time to complete the complete workflow            |
| TOOL-10 | Incorrect/unsafe tool invocation rate | %     | Attempts to invoke inappropriate tools/actions    |

# Experimental Methodology Requirements

## Controlled Variables

To ensure fair comparisons, the following variables must be documented and controlled where applicable:

- Hardware configuration and available memory.
- Operating system and virtualization configuration.
- Docker and inference runtime versions.
- Model name, version, parameter count, and quantization.
- Context window and maximum output tokens.
- Temperature, sampling parameters, and random seed where supported.
- Benchmark dataset and prompt versions.
- Concurrency level and background workloads.
- Model warm-up and caching conditions.

If settings cannot be made equivalent across models, differences must be documented.

## Benchmark Repetitions

Each benchmark task should initially be executed at least three times per model configuration.

Additional repetitions may be required for tasks exhibiting high variability or unreliable outcomes.

Results must include individual execution records as well as aggregate statistics, such as mean, median, and relevant percentiles.

## Model Selection

Candidate models will be selected using the following criteria:

- Availability of model weights for local execution.
- Compatibility with the selected inference runtime.
- Parameter size and expected hardware requirements.
- Suitability for the target use cases.
- Quantization availability.
- Context window support.
- License and potential commercial-use restrictions.
- Available documentation and model maintenance.

The initial evaluation will consider approximately 1–3B, 4–5B, 7–9B, and 12–14B parameter tiers, with larger models used selectively for hardware-limit investigation.

## Data Storage

The benchmark database must retain:

- Unique experiment and execution identifiers.
- Model and configuration information.
- Benchmark dataset and task identifiers.
- Timestamps and execution duration.
- Raw outputs and expected outputs.
- Hardware measurements.
- Evaluation scores.
- Execution status and error information.

The database design must allow multiple runs of the same model and task without overwriting previous results.

## Validation

Before full-scale benchmarking, the measurement system must be validated using a small pilot experiment.

The pilot must verify that:

1. The model receives the expected prompts.
2. Responses are correctly captured.
3. Timing and resource metrics are recorded accurately within documented measurement limitations.
4. Failed requests are stored.
5. Results can be retrieved and compared from the database.
6. The inference pipeline works without external AI services.

## Feasibility Assessment

The benchmark results will support a feasibility assessment across four dimensions.

| Dimension               | Assessment                                                                             |
|-------------------------|----------------------------------------------------------------------------------------|
| Technical feasibility   | Whether the model can execute successfully on the allocated hardware.                  |
| Functional feasibility  | Whether the model achieves acceptable results for the target use cases.                |
| Operational feasibility | Whether performance, memory consumption, and reliability support practical deployment. |
| Financial feasibility   | Whether potential benefits justify infrastructure and operating costs.                 |